MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC TEST – 02 (POLYNOMIALS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective & MCQs

MATHS MANTHAN | $5 \times 1 = 5$ Marks

Q1. What should be subtracted from the polynomial $x^2 - 16x + 30$ so that 15 is a zero of the resulting polynomial? [1]

(A) 30 (B) 14 (C) 15 (D) 16

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Let $p(x) = x^2 - 16x + 30$. Subtracting k gives $q(x) = p(x) - k$. We know:

Formula Used: 15 is a zero of $q(x) \implies q(15) = 0 \implies k = p(15)$

Evaluating $p(15)$:

$$p(15) = 15^2 - 16(15) + 30 = 225 - 240 + 30 = 15 \implies \boxed{k = 15}. \quad [1M]$$

Q2. The zeroes of the quadratic polynomial $x^2 + ax + a$, $a > 0$: [1]

(A) cannot both be positive **(B)** cannot both be negative

(C) are always unequal **(D)** are always equal

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Let α, β be the zeroes of $x^2 + ax + a$ $(a > 0)$. We know Vieta's relations:

Formula Used: $\alpha + \beta = -a < 0, \qquad \alpha\beta = a > 0$

Since $a > 0$, the sum of zeroes is $-a < 0$. Two positive numbers can never add up to a negative number, so the zeroes $\boxed{\text{(A) cannot both be positive}}$. [1M]

Q3. The value of x for which the polynomial $p(x) = x^2 - 2x - 8$ has a minimum value is: [1]

(A) 1 (B) -1 (C) 2 (D) -4

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

We use the Class 10 completing-the-square form of $p(x) = x^2 - 2x - 8$:

Formula Used: $x^2 - 2x - 8 = (x-1)^2 - 9$

Since $(x-1)^2 \geq 0$ for all real x with equality iff $x - 1 = 0$, the minimum value -9 occurs at $\boxed{x = 1}$. [1M]

Q4. If one zero of the quadratic polynomial $4x^2 - 8kx - 9$ is negative of the other, then $k =$: [1]

(A) 1 (B) -1 (C) 0 (D) 2

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (C)

Let the zeroes be α and $-\alpha$. We know Vieta's sum formula:

Formula Used: Sum of zeroes $= \alpha + (-\alpha) = 0 \implies -\dfrac{b}{a} = 0$

Here $a = 4$, $b = -8k$, so:

$$-\frac{(-8k)}{4} = 0 \implies 2k = 0 \implies \boxed{k = 0}. \qquad [1M]$$

Q5. Assertion-Reason: [1]

Assertion (A): If the sum and product of zeroes of a quadratic polynomial are -3 and 2 respectively, then the polynomial is $x^2 + 3x + 2$.

Reason (R): A quadratic polynomial with sum S and product P is given by $k(x^2 - Sx + P)$.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) Both A and R are true and R is the correct explanation of A.

We know the construction formula given in Reason **(R)**:

Formula Used: $p(x) = k(x^2 - Sx + P)$

Substituting $S = -3$, $P = 2$, and taking $k = 1$ gives $p(x) = x^2 - (-3)x + 2 = x^2 + 3x + 2$. Thus both **(A)** and **(R)** are true, and **(R)** is the correct explanation of **(A)**. [1M]

SECTION B: Very Short Answer Questions — MATHS MANTHAN | 2 × 2 = 4 Marks

Q6. If α and β are the zeroes of the polynomial $x^2 - 4x + 3$, find the value of $\alpha^4\beta^3 + \alpha^3\beta^4$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

From $x^2 - 4x + 3$, we have $a = 1$, $b = -4$, $c = 3$. We know Vieta's relations:

Formula Used: $\alpha + \beta = -\dfrac{b}{a} = 4, \qquad \alpha\beta = \dfrac{c}{a} = 3$

Factoring out the common term $\alpha^3\beta^3 = (\alpha\beta)^3$:

$$\alpha^4\beta^3 + \alpha^3\beta^4 = \alpha^3\beta^3(\alpha + \beta) = (\alpha\beta)^3(\alpha + \beta) \quad [1M] = (3)^3(4) = 27 \times 4 = \boxed{108}. \quad [1M]$$

Q7. Find a quadratic polynomial whose zeroes are reciprocals of the zeroes of $f(x) = ax^2 + bx + c$, $a \neq 0$, $c \neq 0$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let α, β be the zeroes of $f(x) = ax^2+bx+c$. Since $c \neq 0$, neither zero is 0, and Vieta gives $\alpha+\beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$. We know the polynomial form for new zeroes $\frac{1}{\alpha}, \frac{1}{\beta}$:

Formula Used: $g(x) = k\left(x^2 - S'x + P'\right)$

Computing S' and P':

$$S' = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-b/a}{c/a} = -\frac{b}{c}, \qquad P' = \frac{1}{\alpha} \cdot \frac{1}{\beta} = \frac{1}{\alpha\beta} = \frac{a}{c}. \quad [1M]$$

Substituting into $g(x)$ and choosing $k = c$ gives $\boxed{g(x) = cx^2 + bx + a}$ (or $k(cx^2 + bx + a)$). **[1M]**

SECTION C: Short Answer Questions — MATHS MANTHAN | 2 × 3 = 6 Marks

Q8. If α and β are zeroes of $p(x) = 3x^2 + 2x + 1$, find the polynomial whose zeroes are $\dfrac{1-\alpha}{1+\alpha}$ and $\dfrac{1-\beta}{1+\beta}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

From $p(x) = 3x^2+2x+1$, $\alpha+\beta = -\frac{2}{3}$ and $\alpha\beta = \frac{1}{3}$ (since $p(-1) = 3-2+1 = 2 \neq 0$, $1+\alpha \neq 0, 1+\beta \neq 0$). We know:

Formula Used: $q(x) = k\left[x^2 - S'x + P'\right]$

First evaluate the common denominator $D = (1+\alpha)(1+\beta) = 1 + (\alpha+\beta) + \alpha\beta = 1 - \frac{2}{3} + \frac{1}{3} = \frac{2}{3}$. **[0.5M]**

Now compute S' and P':

$$S' = \frac{1-\alpha}{1+\alpha} + \frac{1-\beta}{1+\beta} = \frac{(1-\alpha)(1+\beta) + (1-\beta)(1+\alpha)}{(1+\alpha)(1+\beta)} = \frac{2 - 2\alpha\beta}{\frac{2}{3}} = \frac{2 - \frac{2}{3}}{\frac{2}{3}} = \frac{4/3}{2/3} = 2, \quad [1M]$$

$$P' = \frac{(1-\alpha)(1-\beta)}{(1+\alpha)(1+\beta)} = \frac{1 - (\alpha+\beta) + \alpha\beta}{\frac{2}{3}} = \frac{1 + \frac{2}{3} + \frac{1}{3}}{\frac{2}{3}} = \frac{2}{2/3} = 3. \quad [1M]$$

Choosing $k = 1$, the required polynomial is $\boxed{q(x) = x^2 - 2x + 3}$. **[0.5M]**

Q9. If α and β are the zeroes of $f(x) = x^2 - 1$, find a quadratic polynomial whose zeroes are $\dfrac{2\alpha}{\beta}$ and $\dfrac{2\beta}{\alpha}$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

From $f(x) = x^2 - 1 = (x-1)(x+1)$, we have $\alpha + \beta = 0$ and $\alpha\beta = -1$ (the zeroes are 1 and -1). We know:

Formula Used: $g(x) = k\big[x^2 - S'x + P'\big], \qquad \alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$

Calculating S' and P' for $\alpha' = \frac{2\alpha}{\beta}$ and $\beta' = \frac{2\beta}{\alpha}$:

$$S' = \frac{2\alpha}{\beta} + \frac{2\beta}{\alpha} = \frac{2(\alpha^2+\beta^2)}{\alpha\beta} = \frac{2\big[(\alpha+\beta)^2 - 2\alpha\beta\big]}{\alpha\beta} = \frac{2\big[0^2 - 2(-1)\big]}{-1} = \frac{4}{-1} = -4, \quad \textbf{[1M]}$$

$$P' = \left(\frac{2\alpha}{\beta}\right)\left(\frac{2\beta}{\alpha}\right) = 4. \quad \textbf{[1M]}$$

Taking $k = 1$, the required quadratic polynomial is $g(x) = x^2 - (-4)x + 4 = \boxed{x^2 + 4x + 4}$. **[1M]**

SECTION D: Long Answer / HOTS Questions — MATHS MANTHAN | 2 × 5 = 10 Marks

Q10. If α and β are the zeroes of $f(x) = 2x^2 - 5x + 7$, form a quadratic polynomial whose zeroes are $\dfrac{\alpha}{\beta}$ and $\dfrac{\beta}{\alpha}$. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Comparing $f(x) = 2x^2 - 5x + 7$ with $ax^2 + bx + c$, we obtain:

$$\alpha + \beta = \frac{5}{2}, \qquad \alpha\beta = \frac{7}{2}. \quad \textbf{[1M]}$$

We know the construction formula and the symmetric sum-of-squares identity:

Formula Used: $g(x) = k\big[x^2 - S'x + P'\big], \qquad \alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$

Computing $\alpha^2 + \beta^2$:

$$\alpha^2 + \beta^2 = \left(\frac{5}{2}\right)^2 - 2\left(\frac{7}{2}\right) = \frac{25}{4} - 7 = \frac{25-28}{4} = -\frac{3}{4}. \quad \textbf{[1M]}$$

Now compute the new sum S' and product P':

$$S' = \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{-3/4}{7/2} = -\frac{3}{4} \times \frac{2}{7} = -\frac{3}{14}, \quad \textbf{[1.5M]}$$

$$P' = \left(\frac{\alpha}{\beta}\right)\left(\frac{\beta}{\alpha}\right) = 1. \quad \textbf{[0.5M]}$$

Substituting into $g(x)$ and choosing $k = 14$ gives $g(x) = 14\left(x^2 + \frac{3}{14}x + 1\right) = \boxed{14x^2 + 3x + 14}$. **[1M]**

Q11. If α and β are the zeroes of $f(x) = x^2 - 2x + 3$, find a quadratic polynomial whose zeroes are: **(i)** $\dfrac{\alpha-1}{\alpha+1}$ and $\dfrac{\beta-1}{\beta+1}$ **(ii)** α^3 and β^3. [5]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

From $f(x) = x^2 - 2x + 3$, we have $\alpha + \beta = 2$ and $\alpha\beta = 3$ ($D = -8 < 0$, so symmetric algebraic Vieta relations hold over $\mathbb{C}$). We know:

Formula Used: $g(x) = k\big(x^2 - S'x + P'\big)$, $\quad \alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$

(i) Zeroes $\dfrac{\alpha-1}{\alpha+1}$ and $\dfrac{\beta-1}{\beta+1}$: Note $(\alpha+1)(\beta+1) = \alpha\beta + (\alpha+\beta) + 1 = 3 + 2 + 1 = 6$. [0.5M]

$$S' = \frac{\alpha-1}{\alpha+1} + \frac{\beta-1}{\beta+1} = \frac{(\alpha-1)(\beta+1) + (\beta-1)(\alpha+1)}{6} = \frac{2\alpha\beta - 2}{6} = \frac{2(3)-2}{6} = \frac{4}{6} = \frac{2}{3},$$

$$P' = \frac{(\alpha-1)(\beta-1)}{(\alpha+1)(\beta+1)} = \frac{\alpha\beta - (\alpha+\beta) + 1}{6} = \frac{3-2+1}{6} = \frac{2}{6} = \frac{1}{3}. \quad [1.5\text{M}]$$

Setting $k = 3$ yields $g_1(x) = 3\left(x^2 - \frac{2}{3}x + \frac{1}{3}\right) = \boxed{3x^2 - 2x + 1}$. [0.5M]

(ii) Zeroes α^3 and β^3:

$$S' = \alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta) = (2)^3 - 3(3)(2) = 8 - 18 = -10, \quad [1\text{M}]$$

$$P' = \alpha^3\beta^3 = (\alpha\beta)^3 = 3^3 = 27. \quad [1\text{M}]$$

Setting $k = 1$ yields $g_2(x) = x^2 - (-10)x + 27 = \boxed{x^2 + 10x + 27}$. [0.5M]

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Q12. Case Study: Mountain Railway Tunnel Arch

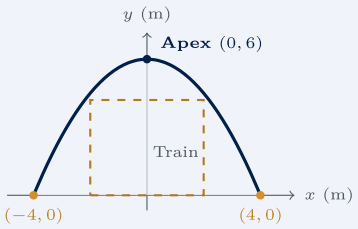
A single-track railway tunnel has a parabolic entrance of width 8 m at ground level and maximum height 6 m at the centre, taking the centre of the track on the ground as $(0, 0)$. [5]

(i) Write the coordinates of the two ground contact points of the arch and the apex point. **[1]**

(ii) Form the quadratic polynomial $y = f(x)$ defining the tunnel entrance arch profile. **[2]**

(iii) **(a)** Find the height of the tunnel arch at a distance of 2 m from the centre line. **OR (b)** A rectangular freight container train of width 4 m (2 m on each side) and height 4.2 m enters the tunnel. Determine with calculations if it will pass safely. **[2]**

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Cross-Section Geometry: Symmetric parabola anchored at $(-4, 0)$ and $(4, 0)$ with peak 6 m at $x = 0$. Dashed rectangle illustrates the 4 m × 4.2 m freight container inside $y(\pm 2) = 4.5$ m.

Formula Used: $y = a(x - \alpha)(x - \beta)$

(i) Key coordinates: Because total base width 8 m is symmetric about the y-axis through $(0, 0)$, the ground contacts extend 4 m on each side:

$$\text{Ground contacts: } \boxed{(-4, 0) \text{ and } (4, 0)}, \qquad \text{Apex: } \boxed{(0, 6)}. \qquad [1M]$$

(ii) Arch polynomial $y = f(x)$: Using zeroes $\alpha = -4$ and $\beta = 4$:

$$f(x) = a(x + 4)(x - 4) = a(x^2 - 16). \qquad [1M]$$

Since the arch passes through apex $(0, 6)$, substitute $x = 0, y = 6$:

$$a(0 - 16) = 6 \implies a = -\frac{6}{16} = -\frac{3}{8} \implies \boxed{f(x) = -\frac{3}{8}x^2 + 6}. \qquad [1M]$$

(iii)(a) Height at $x = 2$ m: Substituting $x = 2$:

$$f(2) = -\frac{3}{8}(2)^2 + 6 = -\frac{12}{8} + 6 = -1.5 + 6 = \boxed{4.5 \text{ m}}. \qquad [2M]$$

OR (Alternative Part (b) – Train clearance check): A container train of width 4 m centered along $x = 0$ spans $x \in [-2, 2]$. Minimum clearance over $[-2, 2]$ occurs at edges $x = \pm 2$:

$$f(\pm 2) = -\frac{3}{8}(4) + 6 = -1.5 + 6 = \boxed{4.5 \text{ m}}. \qquad [1M]$$

Comparing with train height 4.2 m: 4.5 m $>$ 4.2 m (clearance margin = 0.3 m at both edges). Therefore, $\boxed{\text{Yes, the container train passes safely.}}$ [1M]

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